Spirulina Cultivation: A New Source of Income for Farmers

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Abstract

In today's world, where great importance is being given to healthy food, Spirulina, recognized as a "super food", offers a wonderful opportunity in the agricultural sector. This blue-green microalga, which can be grown in small ponds or tanks, is rich in proteins, vitamins, and minerals and provides several health benefits. Requiring less land, less water, and low investment while offering high returns, spirulina cultivation is opening new avenues for farmers and young entrepreneurs. In the southern states of India, this enterprise is gradually gaining momentum, with many young farmers coming forward to adopt it. However, developing a proper marketing channel is a very crucial aspect of the spirulina business.

What is Spirulina?

Spirulina is a type of microalga. It is blue-green in color and is known as a super food due to its high nutritional value.

- Contains about 60-70% protein
- Rich in vitamin B12, iron, beta-carotene, and other vitamins and minerals
- High antioxidant content helps in boosting immunity

Requirements for Spirulina Cultivation

A.Spirulina cultivation is mainly water-based.
B.Tanks/Ponds: Can be grown in cement tanks or plastic-lined ponds (commonly of size 130 × 22 feet).
C.Water and Nutrient Solution: Clean drinking water or borewell water mixed with sodium bicarbonate, urea, super phosphate, sea salt, etc.
D.Sunlight: A location receiving at least 6-8 hours of sunlight per day.
E.pH Level: Alkaline solution with a pH of 9-11.
F.Temperature: Optimum temperature range is 25-35°C.

Method of Spirulina Production

- First, the cultivation site should be properly levelled.
- Construct a cement tank with 130 feet length, 22 feet width, and 1.5 feet depth.
- Spread tarpaulin sheets (44 feet width) over the tanks in such a way that water can be retained properly.
- Maintain a water depth of about 25 cm in these tanks at all times.
- Fill the tanks with water. To convert this water into a seawater-like medium, initially add 100 kg of salt.
- For regular maintenance, add common baking soda (used in household cooking) to the water. Additionally, add neem oil, magnesium, and sulphur as required.
- Install a 1 HP rotator in the water tanks to continuously mix the water. This ensures uniform temperature throughout the tank and supports healthy spirulina growth.
- About 70 kg of mother culture should be added to one water tank.
- Production starts after 15 days of adding the culture.
- Once production begins, one batch of spirulina is ready every 24 hours.
- Using a bucket arrangement and a water suction pump, spirulina-rich water from the main tank is passed through a nylon cloth into a separate small tank.

- The spirulina biomass gets collected on the nylon cloth.
- The collected spirulina is transferred into separate buckets and dried properly.
- In this way, spirulina is processed step by step into a usable product.

Cost of Production and Returns in Spirulina Cultivation
Costs
- Construction of 2 tanks (one unit): ₹2-3 lakhs
- Cost to produce 1 kg of dry spirulina powder: ₹100-200 per kg
- Monthly maintenance cost: ₹20,000-₹30,000, (labour, nutrient solution, water testing, etc.)

Returns
- 60 kg of wet spirulina per unit per day
- From this, 5-6 kg of dry spirulina powder per day is obtained
- Market price of dry spirulina: ₹500 per kg
- Net profit per kg after expenses: ₹300
- Daily net profit:
- 5 kg × ₹300 = ₹1,500 per day
- Monthly net profit:
- ₹1,500 × 30 days = ₹45,000 per month

Market and Uses of Spirulina
1.Health food industry - Spirulina is widely used to manufacture tablets, capsules, and protein powders.

2.Bakery and food products - It is increasingly used in cookies, pasta, soups, and juices.

3.Animal feed and aquaculture - Used as a nutritional supplement in cattle feed, fish feed, and poultry feed, which helps increase milk yield, meat production in poultry, and fish growth.

4.Cosmetics industry - Spirulina is used in skin care and hair care products due to its antioxidant and nourishing properties.

Conclusion
In this era of growing health awareness, spirulina production is a future-oriented enterprise. With low investment and high profitability, it offers excellent opportunities for farmers, youth, and small-scale entrepreneurs. By following proper technology, maintaining hygiene standards, and adopting effective marketing strategies, spirulina cultivation can truly become “green gold.”

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